

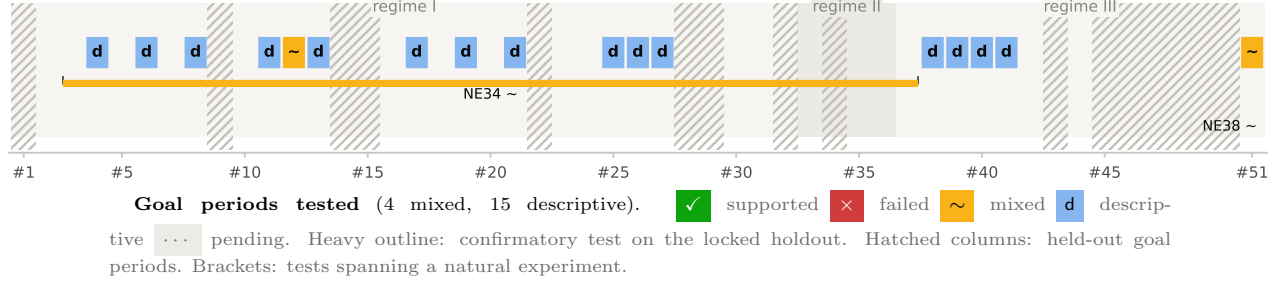
H105: Goals act on a two-state order parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1/5

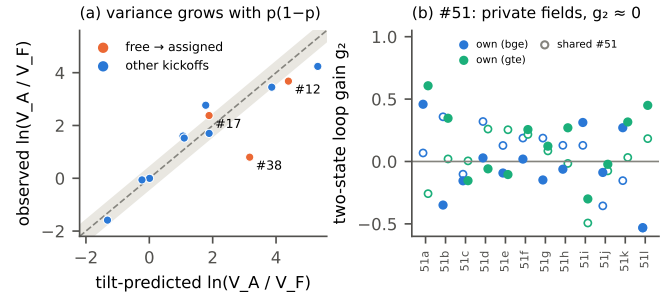
Bottom line. Goals act on a two-state occupancy whose variance grows as $p(1-p)$, but the free-week tilt forecast is not identifiable at village sampling, so the Legendre question stays open.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H10's Legendre test failed on continuous alignment. HH130 proposed a fairer test: treat each agent-window as on-goal or off-goal and ask whether a uniform field (tilt) on the free week predicts the assigned week's occupancy variance and who moves. I ran it on three free → assigned pairs, 16 kickoffs and three natives, in two embedding models. The synthetic came first and showed the tilt test is not identifiable at village sampling: with a true tilt it passes in 30% of pairs, and in 40% under a coupling change. The two-state picture holds descriptively. Occupancy rises from the decoy floor (0.02–0.05) to 0.23–0.44, its variance grows as $p(1-p)$ (Spearman 0.86 over 16 kickoffs), and transverse occupancy stays flat. Who moves: inconclusive in 3/3 pairs. Holdout: not run.



Observables. (a) Observed against tilt-predicted growth of the occupancy variance, free (or previous) week to assigned week, for 16 kickoffs; the band is $\pm \ln 1.5$. The ranks agree ($\rho_s = 0.86$), but only 7/16 fall in the band; #38a sits far below. (b) #51: loop gain of each agent's spin along its own private goal (filled) and along the shared #51 direction (open), per unit. Private fields do not synchronize, except the role-onset unit 51a.

PHYSICAL MODEL

Model 01, forward mean field (Curie–Weiss with agent fields), in 0/1 spins. Spin $\sigma_{i,t}$: agent i 's statements in 30-min window t are mostly on-goal (projection on $\hat{g}$ above the 95th percentile of decoy statements). Field h_i : each agent's on-goal tendency. Coupling J_2 : synchronized switching. Occupancy p_i : mean spin. The tilt adds one λ to every h_i and keeps J_2 . g_2 : two-state loop gain.

$$P(\sigma) \propto \exp\left(\sum_i h_i \sigma_i + \frac{J_2}{2N} n^2\right), \quad n = \sum_i \sigma_i$$

$$\text{Var}(n) = \frac{S}{1 - J_2 \bar{q}}, \quad S = \sum_i p_i(1 - p_i), \quad g_2 = J_2 \bar{q}$$

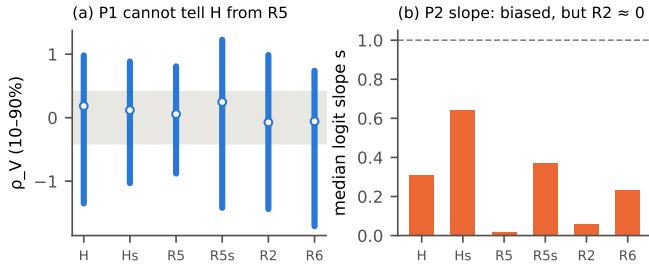
$$\text{tilt: } h_i \rightarrow h_i + \lambda \Rightarrow \rho_V = \ln \frac{V_A^{\text{obs}}}{V_A(\lambda, J_2^F)} \approx 0$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a methods result. The two-state view is the right bookkeeping for a goal: it moves only the on-goal occupancy, from about 0.03 to 0.2–0.45, and leaves other directions alone. But the forecast HH130 wanted, steerability from a free week, cannot be made from village logs with this instrument: free weeks have almost no on-goal windows, so their coupling is unmeasurable, and 2–4 statements per window misclassify the spin. An operator learns nothing new beyond H10 and H54. A monitor builder learns that occupancy measured this way needs a calibrated classifier before any fluctuation–response forecast.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 tilt variance	$\rho_V -0.71, +0.53, -2.45$; gte flips #12a	not identifiable
P2 uniform logit shift	$s -0.60, +0.38, +2.53$; calibrated 0/3	inconclusive
P3 J_2 unchanged	$\Delta g_2 +0.64, +0.26, +0.03$	not identifiable
P4 transverse flat	$p_{\perp} \approx 0.02$ in F and A	supported
P6 kickoffs	7/16 in band; $\rho_s 0.86$ ($p 0.0003$)	half / half
N1 G51 private goals	$g_2 -0.08$ (bge), $+0.19$ (gte)	mixed
N2 G12 debate schedule	removes 50% of excess (gte); no excess (bge)	mixed
N3 NE38 one spin	$0 \rightarrow 0.96$; others $+0.01 / +0.05$	mixed



Second figure. Latent Curie-Weiss spins at the real window structure of the three pairs, observed through statements as in the real pipeline (30 replicates per scenario). (a) The tilt (H) and a coupling change (R5) give the same ρ_V spread. (b) The logit slope is biased low under the tilt (0.31) but near 0 under a common target (R2), so P2 keeps some resolution.

GOAL PERIODS IN DETAIL

G04 d descriptive
G06 d descriptive
G08 d descriptive
G11 d descriptive
G12 ~ mixed
G13 d descriptive
G17 d descriptive
G19 d descriptive
G21 d descriptive
G25 d descriptive
+9 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	0	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

"On-goal" is a text-embedding threshold; the 90% and 98% variants shift p but not the conclusions. Spins built from 2-4 statements are often wrong, which drives the non-identifiability. The two models disagree on the sign of ρ_V in #12a and on both natives' second clauses. Free weeks are not field-free. The variance-growth agreement across kickoffs follows from any binary occupancy and does not test the tilt. The on-goal work variant was not run.

ROUND 2

Build a calibrated on-goal classifier with a measured confusion matrix, prove identifiability on the synthetic, and pool free-week couplings hierarchically before retesting the tilt.

- **H105-R1** Fix the instrument, not the model.
- **H105-R2** Pool the free-week coupling hierarchically (exception (d)) across the free weeks of one regime to measure J_2 at all, then predict each assigned week.
- **H105-R3** Separate drive from coupling in #12 with the DQ6 debate schedule as an explicit time-dependent field and a read-out (ledger) lag test for the residual.

H105-two-state-goal-order · generated 2026-10-04 14:07 by infra/summaries/build_summaries.py · card hypotheses/H105-two-state-goal-order/README.md · ratings: H105 agent (suggested) (page 2)